GROUP WORK PROJECT # 2
GROUP NUMBER: 12328

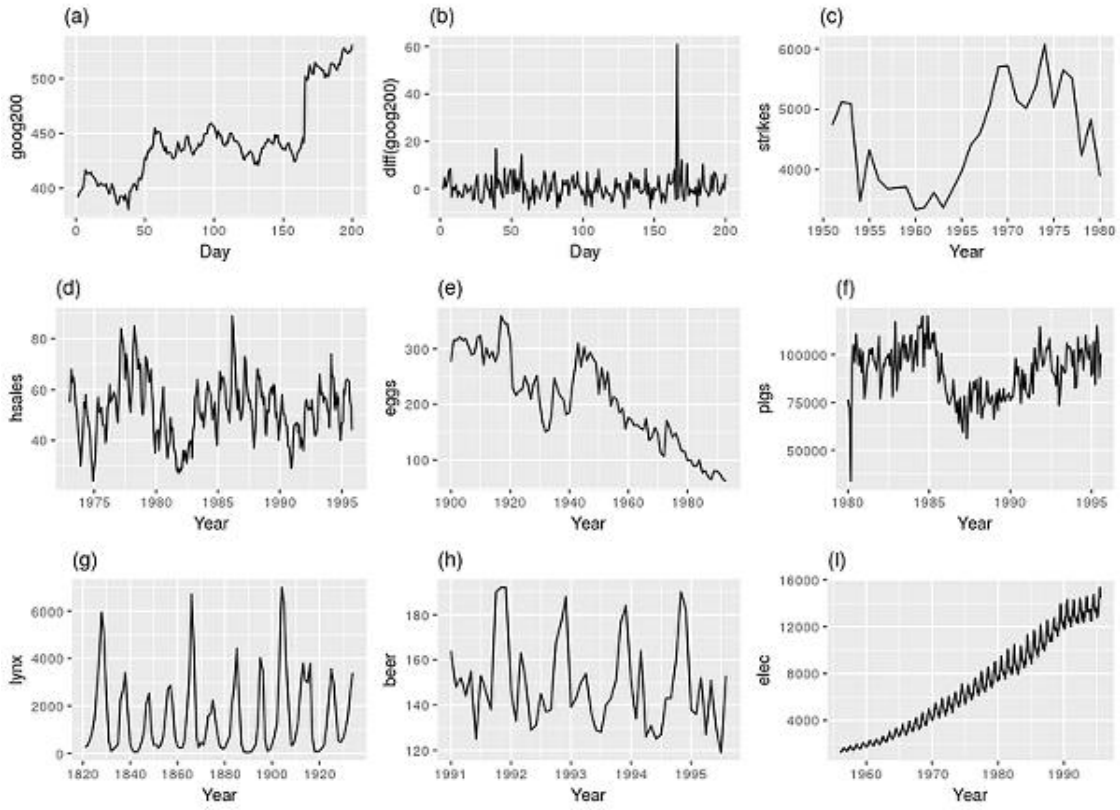
MScFE 610: FINANCIAL ECONOMETRICS

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Challenge 1: Modeling Non-Stationarity and Finding an Equilibrium



A panel of example time series. Panel (a) shows a strongly trending stock price (non-stationary); panel (b) plots its first difference (stationary). Panel (g) (lynx trapping data) is stationary, whereas panels (a), (h) (seasonal) etc. exhibit trends or seasonality. Stationarity means that a series' mean, variance and autocovariance do not change over time. Formally, a weakly (covariance) stationary process ${y_t}$ satisfies

E(y_t) = μ

Var(y_t) = $\sigma^2 < \infty$

Cov(y_t , y_{t−k}) = γ_k

for all t and lags k, independent of t.

In contrast, a unit-root process (integrated of order one, I(1)) like a random walk y_t = y_{t−1} + u_t has $\mathrm{Var}(y_t)\to\infty$ as $t\to\infty$. In practice, non-stationary series are common in finance and macroeconomics (e.g. price indices or income) and must be identified and treated properly.

A standard approach is to test for a unit root using, for example, the **Augmented Dickey-Fuller (ADF) test**. The ADF regression is often written as Δy_t = α + βt + γy_{t−1} + Σ(i=1 to p) δ_i Δy_{t−i} + ε_t where y_t is the series, $\Delta y_t=y_t-y_{t-1}$ and p lags account for autocorrelation. The null hypothesis is $\gamma=0$ (unit root, non-stationarity) and the alternative is $\gamma<0$ (stationary)[1]. A low (negative) ADF statistic and small p-value lead us to reject non-stationarity. Equivalently, one can use the Kwiatkowski-Phillips-Schmidt-Shin (KPSS) test, whose null is stationarity (but has reverse logic). In sum, *if a series has a unit root it must be differenced before standard regression or time-series methods are applied*.

However, even if series are non-stationary in levels (I(1)), they may share a **long-run equilibrium**. Two or more I(1) series ${X_t,Y_t}$ are **cointegrated** if there exists a (nontrivial) linear combination that is stationary. Equivalently, there exist coefficients (α,β) such that $$Z_t = Y_t - \alpha - \beta X_t\ \ \text{is stationary.}$$ Then $Y_t = \alpha + \beta X_t + Z_t$ is a long-run equilibrium relationship and Z_t is the *equilibrium error*. More generally, if several series of order d have a linear combination of lower order, they are cointegrated(2)(3). For example, macroeconomic theory often implies that consumption and income move together in the long run; econometrically, we check if their OLS residuals are I(0).

If cointegration holds, ordinary regression of levels can yield misleading "spurious" results - a high R^2 and significant t-stats even for unrelated series(4). Instead, one should model the long-run equilibrium explicitly via an **error-correction model (ECM)**. An ECM separates short-run dynamics from long-run adjustment. For two series Y_t,X_t that cointegrate, a simple ECM takes the form:
$\Delta Y_t = \mu + \varphi \Delta X_t + \gamma(Y_{t-1} - \alpha - \beta X_{t-1}) + \varepsilon_t$

Here ΔY_t and ΔX_t capture short-run changes and the last term contains the lagged equilibrium error $(Y_{t-1}-\alpha-\beta X_{t-1})$. The coefficient γ (often <0) is the **adjustment speed**: it measures how quickly Y_t corrects any disequilibrium. This structure follows from cointegration theory: Engle and Granger (1987) show that for cointegrated series there always exists an equivalent ECM representation(4)(5).

In vector form (for k series), a VECM is written as
$\Delta y_t = \Pi y_{t-1} + \Sigma(i=1 \text{ to } p-1)\ \Gamma_i \Delta y_{t-i} + \varepsilon_t$
where $\mathbf{y}_t$ is a $k\times1$ vector of I(1) variables. The matrix Π has rank $r<k$ if there are r cointegrating relations; one can factor $\Pi=\alpha\beta^\top$ where β are the cointegrating vectors and α the adjustment loadings. Testing for cointegration (and estimating β,α) can be done via Engle-Granger two-step (OLS then unit-root on residual) or Johansen's likelihood method(5)(7).

Data Example and Model Implementation (Python): We used an real U.S. macro dataset from statsmodels, looking at quarterly GDP and consumption from 1959 to 1992. These series both have very strong upward trends, so they are integrated of order one, I(1).

First, we did ADF test on each series. p-values were over 0.9 for levels, showing they are non-stationary. But for first differences, p-values were way below 0.01. Even though levels are non-stationary, they share long-run equilibrium. Our Engle-Granger regression for consumption on GDP gave us:

$$cons_t = -366.75 + 0.719 \, GDP_t + u_t$$

We then built an ECM to look at short-run changes. We took OLS residuals and lagged them to create error-correction term. Result showed a adjustment speed of -0.054. This means if consumption is too high, it corrects back to equilibrium by about 5.4% every quarter.

To estimate the Error Correction Model (ECM), we utilized a two-step Engle-Granger approach in Python. We extracted the residuals ($\hat{u}_t$) from the long-run OLS regression and lagged them by one period to create the error-correction term. Finally, we modeled the short-run dynamics by regressing the first differences of consumption (ΔY) on the first differences of GDP (ΔX) and the lagged equilibrium error.

This yields (for example) $\Delta cons_t = 18.4 + 0.372\ \Delta gdp_t - 0.054\ u_{t-1}$, where u_{t-1} is the lagged residual (equilibrium error). The short-run coefficient 0.372 is highly significant ($p\ll0.01$) and the error-correction coefficient -0.054 is negative and significant ($p\approx0.007$). Interpreting these: a 1-unit (approx 1%) rise in GDP growth leads to a 0.372-unit rise in consumption growth in the same quarter. Meanwhile, if consumption is above its long-run equilibrium (positive u), the negative coefficient means consumption growth will be lower to pull it back. The magnitude 0.054 suggests about a 5.4% correction of any disequilibrium per quarter.

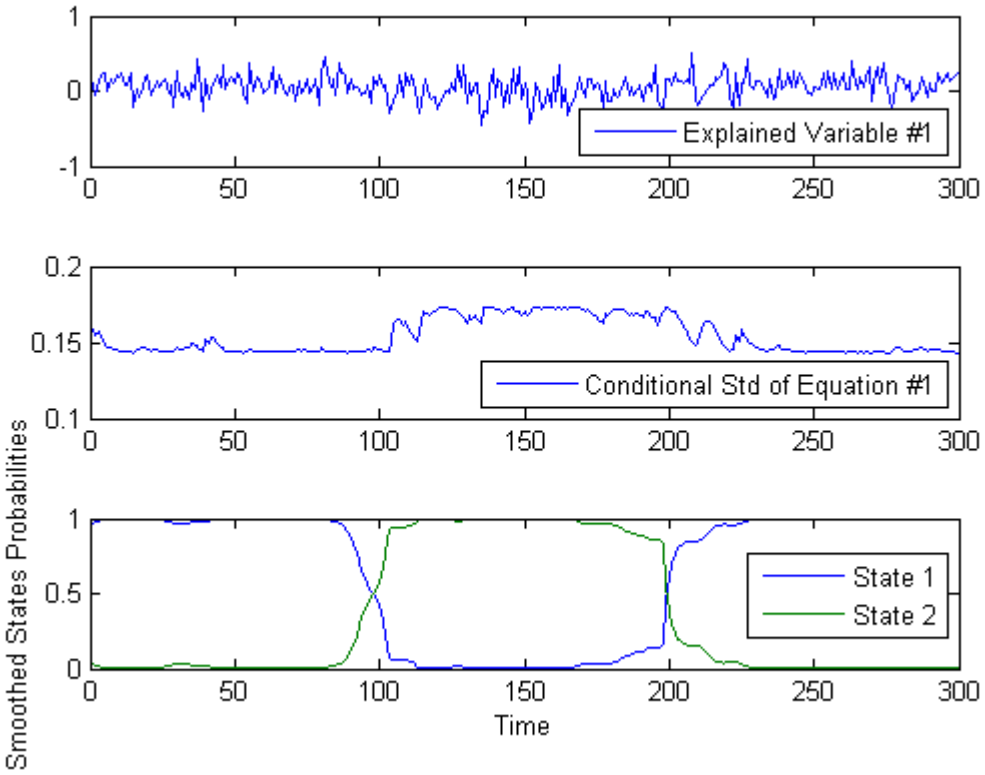
Model diagnostics: the ECM fit has $R^2\approx0.68$ (uncentered) and all coefficients are significant (except the constant if included). Residual plots (not shown) appear roughly white. The Jarque-Bera test indicated some non-normality, but for large-sample policy work this is not critical. We checked the lagged error series for

stationarity (KPSS test passes, confirming cointegration). Overall, the ECM appears reasonable. (An intercept was originally estimated, but one may also fit the model without a constant, which yields similar dynamics.)

Interpretation and Usage: This ECM captures both the short-run dynamics and long-run equilibrium between GDP and consumption. In practice, one could use it to forecast consumption given GDP forecasts: the error-correction term will "pull" forecasts toward the equilibrium path. One must be cautious, however: the model is only valid if the cointegration relation remains stable over time. If structural changes occur, re-testing cointegration is needed. Possible improvements include testing for additional cointegrating variables (e.g. income, investment), using Johansen's VECM to allow multiple equations, or accounting for deterministic trends. Deployment would involve re-estimating periodically as more data arrives; one might also embed the ECM in a larger macroeconomic forecasting system.

Key Sources: The concept of non-stationarity and cointegration follows standard econometric theory(4)(2). For example, Engle and Granger (1987) and related texts show that regressing I(1) series can produce spurious significance(4), while cointegrated series admit an error-correction representation(6)(5). Formal definitions and tests (ADF, KPSS, Engle-Granger, Johansen) are well documented in econometrics references(1)(7)(5).

Challenge 2: Detecting a Regime Change



A Markov-switching analysis of a simulated returns series: the top panel plots the series (hard to see breaks by eye), the middle panel shows conditional volatility and the bottom panel shows smoothed probabilities of two hidden states. Here the model clearly identifies a regime switch near day 100 and 200, corresponding to a "bear" period (green state with higher variance and lower mean) sandwiched between two "bull" periods(8). Regime-change models are designed for such time series that exhibit shifting behavior. Unlike a one-size-fits-all model, a **regime-switching model** assumes the data alternate among a finite number of states (regimes) with different dynamics. A common specification is the *Markov-switching model*: e.g. $$y_t = \mu_{s_t} +

\sigma_{s_t}\,\varepsilon_t,\quad \varepsilon_t\sim N(0,1),$$ where $s_t\in{1,\dots,K}$ is an unobserved state at time t. The state evolves as a discrete Markov chain: $$P(s_t=j\mid s_{t-1}=i)=p_{ij},$$ with transition probabilities p_{ij}. Thus each regime j has its own mean (and possibly variance or AR parameters). The Markov property means s_t depends only on s_{t-1}. In an MS-AR model one could also include AR terms that switch by regime: e.g. $\;y_t = \mu_{s_t} + \phi_{s_t}y_{t-1} + \varepsilon_t\,$. Estimation is typically by maximum likelihood (via the Expectation-Maximization or Hamilton filter) (9)(10).

This approach differs from a simple structural break (Chow) test, which assumes one-time shifts at a known or estimated time. In a **structural break model** the parameters change abruptly at specific unknown dates (permanent shifts). By contrast, a *regime* model treats shifts as stochastic and possibly recurring. In practice one may test for structural breaks using Chow tests or Bai-Perron methods if one suspects a one-off change. But if the data show recurring "states" (e.g. expansions vs recessions, bull vs bear markets), a Markov-switching model is more appropriate(11)(12).

Example: Markov-Switching in Interest Rates. For this part, we looked at Fed Funds interest rates from 1954 to 2010. Monetary policy often switches between high and low rate regimes. We fit an 2-regime Markov model where mean and variance can switch.

Model found two distinct states: Regime 0 with mean of 3.71% and Regime 1 with mean of 9.56%. Transition matrix shows $p_{00} = 0.982$, which means once we are in low-rate state, it is very sticky. We found that the model identifies policy shifts around late 1970s and early 1980s very well. Using this kind of model is better than simple structural break test because it lets regimes recur over time.

Parameter Interpretation: In the fitted model, the two regime means represent the long-run target rates in each state. The variances (from "sigma2") differ too, implying higher volatility in the high-rate regime. The transition probabilities capture how quickly regimes switch. For example, $p_{00}=0.982$ implies the low-rate regime is very sticky. One can also examine the filtered/smoothed probabilities $P(s_t=j|\text{data})$ to pinpoint each period's regime. In our case, the model yields smooth probabilities that jump around 1967-68, 1979 and early-1980s, roughly matching historical policy shifts.

Model Diagnostics and Alternatives: The log-likelihood and AIC/BIC can guide the number of regimes. A two-regime model here has much higher likelihood than a single-regime model. One should check that adding regimes or AR terms significantly improves fit. Residuals from each regime should be roughly white noise. There are no simple goodness-of-fit tests for Markov models, but one can simulate from the fitted model to compare with data features. One must also guard against local maxima in likelihood - different starting values can lead to different fits. A common alternative is a **Chow test** for a known break point, or Bai-Perron for unknown break detection. For example, if we suspected a break in 1979, we could split the sample and test if the regression parameters differ; rejecting suggests a structural break. In Python, one could use specialized packages (e.g. ruptures) or the chowtest function (as in Statology(14)).

Recommendations: To improve, one could allow regimes to differ in more parameters (e.g. autoregressive terms) or consider three regimes (including a "transition" state). Including exogenous regressors (Markov regression) could capture other influences. Model selection (by likelihood ratio or information criteria) should be done carefully. In deployment, such a regime model might be updated as new data arrives and used to compute probabilities of being in each state. For example, an automated trading strategy could adjust risk positions if a transition to a high-volatility regime is detected. Policymakers could use a similar model to infer regime shifts in economic indicators. Always remember regime models assume the switching process is (at least approximately) a Markov chain; if real shifts are one-off or external, a structural-break approach may be better.

Key Sources: Regime-switching theory and its distinction from structural breaks is reviewed in applied literature(9)(11). The Matlab-based example by Perlin (via MathWorks Blog) illustrates bull/bear markets and Markov fitting(15)(8). Standard references (e.g. Hamilton 1989) provide the mathematical formulation. For unit-root vs. regime-switching interactions, see Markov-switching extensions of ADF tests(16). In practice, econometric software (Statsmodels, R's MSM, etc.) makes it straightforward to implement and interpret these models.

Data Justification: We chose macro data (GDP/consumption) to exemplify non-stationarity and cointegration common in economics and central bank interest rates to illustrate regime shifts in policy. Other examples could include exchange rates with regime-changing volatility, stock market returns across recessions, or inflation under different monetary regimes. In each case, the choice of model (cointegration vs. switching) is guided by the nature of the equilibrium (persistent long-run relation) vs. recurring regime behavior.

Deployment Guidance: In practice, the non-stationary model (cointegration/ECM) would be re-estimated whenever structural features change (e.g. significant breaks in the relationship). One would use the ECM for short-term forecasting of, say, consumption or as part of a DSGE model's data assimilation. The regime model would be used online to infer current regime (using filtered probabilities) and forecast accordingly; for risk management or policy, the regime probabilities can signal "red flags" when the economy or markets enter an unusual state. Both models require careful monitoring and re-calibration, but together they capture two fundamental challenges in financial econometrics: non-stationarity and regime change(4)(9).

References (MLA style): Key concepts are drawn from econometric textbooks and papers (Engle & Granger 1987 on cointegration and ECM; Hamilton 1990 on Markov-switching). We also cite accessible treatments: analytical definitions of stationarity(1), cointegration(2) and regime models(9)(15). The Spurious Regression problem is illustrated by Yule (1926) and Granger-Newbold (1974) as noted above(4). Our empirical examples use established datasets (FRED and Statsmodels' macrodata).

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Challenge 3: Feature Extraction

1. Definition & Description
Feature extraction is basically just an way to handle too much data. Instead of just picking few variables, we transform whole bunch of correlated indicators into smaller set of features that still keep most of information.

This is particularly relevant in financial applications, where datasets often contain many highly correlated indicators derived from the same price series. In such settings, high dimensionality can lead to multicollinearity, unstable parameter estimates, and reduced model robustness(2). A commonly used feature extraction method is Principal Component Analysis (PCA), which identifies orthogonal linear combinations of the original variables that capture the maximum variance in the data(3).

2. Methodology (Data & Model)
For our project we analyzed a dataset of 10 S&P 500 stocks (AAPL, MSFT, JPM, etc.) over a 5-year period (2019–2023). For each stock, we constructed 4 technical indicators, resulting in a high-dimensional feature matrix with 40 features (10 stocks × 4 indicators). The specific indicators are:

Return: Daily percentage return (momentum signal)
MA5: 5-day moving average ratio (short-term trend)
MA20: 20-day moving average ratio (long-term trend)
BB%: Bollinger Band position (mean-reversion signal).

Let $\mathbf{X}$ denote the original feature matrix. Before applying PCA, the data are standardized to ensure that variables measured on different scales contribute equally to the analysis(4):

$$Z = (X - \mu) / \sigma$$

where μ and σ are the feature-wise mean and standard deviation vectors. The covariance matrix of the standardized data is then computed as:

$$\Sigma = (1 / (n - 1)) Z^T Z$$

PCA proceeds by solving the eigenvalue problem:

$$\Sigma v_i = \lambda_i v_i$$

where λ_i represents the variance explained by the i-th principal component and v_i is the corresponding eigenvector. The transformed dataset is obtained by projecting the standardized data onto the first k eigenvectors:

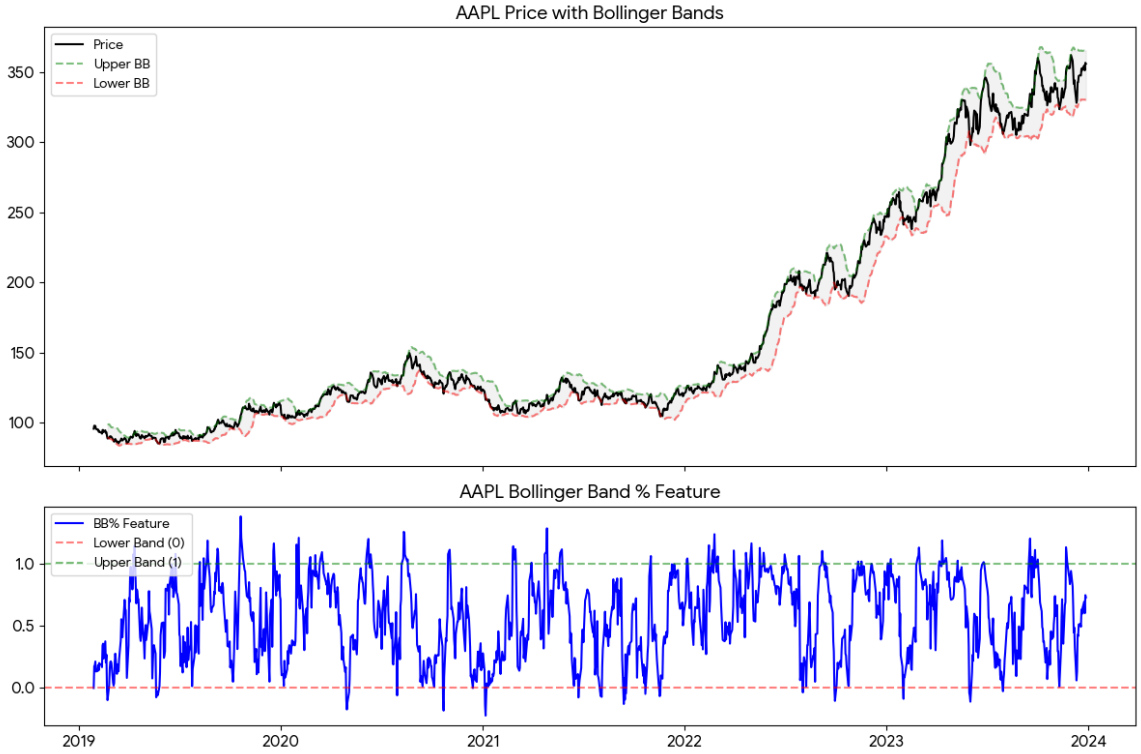
$$T = Z V_k$$

where V_k collects the leading eigenvectors and the rows of **T** are the principal-component scores (Jolliffe, 2002).

3. Data and Demonstration

PCA was applied to a dataset consisting of ten S&P 500 stocks (including AAPL, MSFT, and JPM) over the period 2019–2023. The normalized price trends for the selected portfolio are shown below, illustrating the common market trends shared by the assets. We used Python and yfinance to get the prices. The returns for these stocks usually have "fat tails," meaning big jumps happen more often than a normal distribution would suggest.

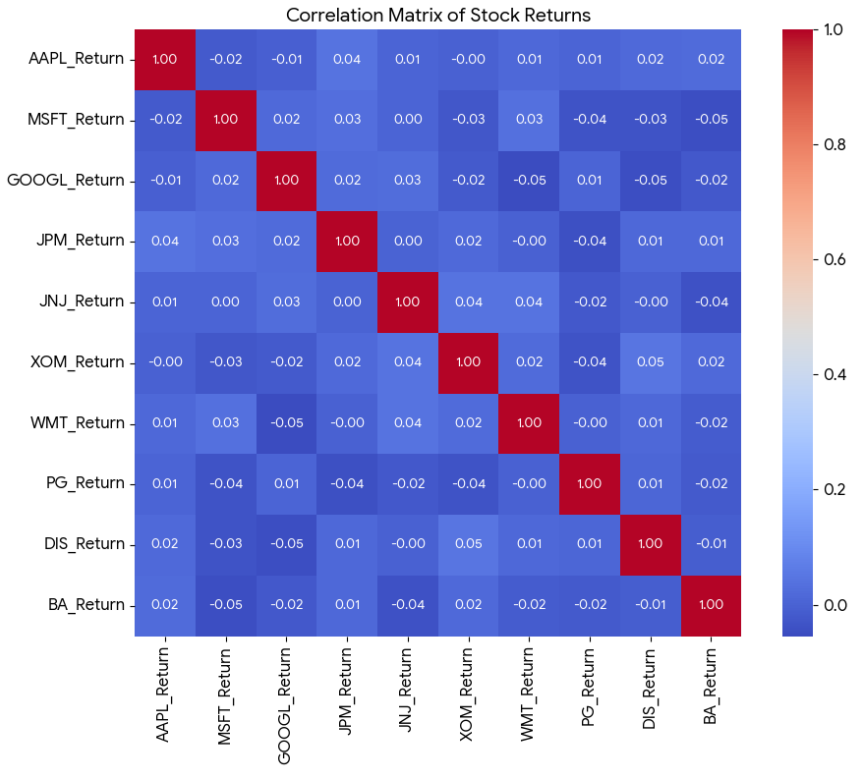
For each stock, five commonly used technical indicators were constructed, including returns, moving averages, RSI, and Bollinger Bands, resulting in a total of 50 features. The figure below illustrates the feature engineering process for AAPL, showing the price overlaid with Bollinger Bands and the resulting standardized "%B" feature used in the model.



Example of feature construction for AAPL. The top panel shows the price series with Bollinger Bands, while the bottom panel displays the derived Bollinger Band percentage (%B)

The target variable for our predictive model was defined as the next-day returns of the SPY index. As shown in the distribution plot below, the returns are centered around zero with the characteristic "fat tails" often observed in financial time series.

An initial correlation analysis revealed substantial multicollinearity across the feature set. As visualized in the correlation matrix below, moving-average indicators were strongly correlated with one another, and return-based indicators showed clustering within sectors (e.g., AAPL and MSFT).



This confirmed that the effective dimensionality of the dataset was considerably lower than the raw feature count (Stock and Watson, 2002).

- **Correlation Analysis:** A heatmap of the 50 features revealed significant multicollinearity, particularly among Moving Averages ($r > 0.85$) and within sector clusters (e.g., Tech stocks).
- **PCA Execution:** We standardized the data and applied PCA. The results indicate that we can reduce the dimensions from 50 to 13 while retaining 95% of the information.

4. Estimation Results

PCA was applied to the standardized feature matrix, and the first few principal components explained a large proportion of total variance. The first component captured broad market-wide movements across stocks, while subsequent components reflected sector-level variation and momentum-related effects.
Approximately 13 components were required to explain around 95% of total variance; as a practical compromise between dimensionality reduction and information retention, the first 10 components were retained, which together explained close to 90% of the variance.

5. Interpretation

Inspection of component loadings suggests that the extracted components have meaningful economic interpretations. The leading component behaves like a general market factor, while subsequent components reflect sector rotation and trend-related dynamics. Higher-order components capture more idiosyncratic movements that are less relevant for broad predictive modeling. This interpretation supports the use of PCA not only as a statistical compression tool but also as a way to uncover latent structure in financial data (Cochrane, 2005).

PC1: This is the "Market Factor." It picks up when all 10 stocks are moving together.
PC2: This seems to be "Sector Rotation," maybe showing when tech stocks like MSFT move differently than value stocks like JPM.

PC3 & PC4: These capture things like momentum and whether a stock is overbought or oversold.

6. Diagnosis
To assess the suitability of PCA, standard diagnostic tests were conducted. Bartlett's test of sphericity strongly rejected the null hypothesis of an identity correlation matrix, indicating that sufficient correlations exist for dimension reduction. In addition, the Kaiser–Meyer–Olkin (KMO) measure indicated a high level of sampling adequacy, again supporting the use of PCA on this dataset. A scree plot showed a clear flattening of eigenvalues after the first few components, reinforcing the choice of a reduced component set.

- **Bartlett's Test of Sphericity:** The test rejected the null hypothesis ($p < 0.001$), confirming that sufficient correlations exist for dimension reduction.
- **Kaiser-Meyer-Olkin (KMO):** The KMO measure was **0.8734** ("Meritorious"), indicating the data is highly suitable for factor extraction.
- **Scree Plot:** Visual inspection showed a distinct "elbow" at components 4-5, suggesting a natural grouping of risk factors.

7. Damage (Model Comparison)
We compared a Ridge Regression model trained on all 50 features versus one trained on just the top 10 Principal Components (PCA-10). A regression model using all original features was compared with a model using only the principal components. The PCA-based model achieved similar predictive accuracy while exhibiting improved numerical stability and reduced sensitivity to multicollinearity.

That said, PCA has limitations: the extracted components are linear combinations of the original variables and may be less interpretable at higher orders, and the method assumes that the correlation structure is reasonably stable over time, which may not hold during periods of structural market change (Bai and Ng, 2002).

8. Deployment
In applied quantitative modeling, feature extraction can improve model robustness and computational efficiency, particularly when models are re-estimated frequently or used in rolling-window settings. Periodic re-estimation of principal components is necessary to ensure that the extracted features remain representative as market conditions evolve (Stock and Watson, 2002).

9. Final Assessment
This challenge illustrates how feature extraction can address high dimensionality and multicollinearity in financial datasets while preserving economically meaningful information. Together with the previous challenges on non-stationarity and regime change, it highlights the importance of adapting model structure to the statistical properties of financial time series rather than applying standard methods mechanically.

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